

The CPU

8.1.3 · Do Now · Objectives

Name: _____ Class: _____

DO NOW · 5 MINUTES

1. True or false: RAM stores start-up programs.

2. What does the subject line mean in an email?

3. Which memory stores data and programs currently running?

4. Which memory stores a program to start up the computer?

LESSON OBJECTIVES

- 1 Name the parts of the CPU using the acronyms.
- 2 Describe the ways you can increase the performance of the CPU.
- 3 Explain why each method of CPU performance makes the CPU quicker.

KEY WORDS

CPU · CU · ALU · Cache · Cores · Clock Speed · Cycles · Hertz · Instructions

CPU Knowledge

Fetch · Decode · Execute · Components

CENTRAL PROCESSING UNIT

Instructions and data are processed by the CPU. CPU is an acronym for Central Processing Unit.

FETCH-DECODE-EXECUTE

1 · FETCH

Fetch instructions from memory.

2 · DECODE

Decode the instruction.

3 · EXECUTE

Execute the instruction.

CPU COMPONENTS

CU · Control Unit

Controls the movement of data around your CPU during the fetch-decode-execute cycle.

ALU · Arithmetic Logic Unit

Carries out all the calculations in the CPU.

Cache

Stores currently used data. It is closer to the CPU so instructions can be fetched faster.

Task 1

Recreate and label the CPU

RECREATE THE CPU PICTURE

Use the original slide as your reference. Draw or recreate the CPU and label CU, ALU and Cache.

Draw / recreate the CPU diagram here

Label: CU · ALU · Cache

EXTENSION

Add a short description of each part.

CU: _____

ALU: _____

Cache: _____

CPU Performance

Cores · Cache · Clock Speed · Exit

A BETTER PERFORMING CPU WILL...

MORE CORES

Process instructions at the same time as each other.

BIGGER CACHE

Store more data to be fetched quickly.

FASTER CLOCK SPEED

Complete more fetch-decode-execute cycles every second.

KEY PHRASES

at the same time · quick access · cycles per second

Clock Speed: how many instructions the CPU runs each second.

SILENT WORKING · 10 MINUTES

Explain why each factor makes the CPU quicker.

More cores:

Bigger cache:

Faster clock speed:
